

Contact Information

Marcos D. (Danny) Caballero (he/they)

Lappan-Phillips Professor of Math and Science Education

Department of Physics & Astronomy

Department of Computational Mathematics, Science, & Engineering

CREATE for STEM Institute

Michigan State University

Biomedical and Physical Sciences Building, 567 Wilson Rd., Room 1310A, East Lansing, MI 48824

Engineering Building, 428 S Shaw Ln, Room 1515A, East Lansing, MI 48824

Email: caball114@msu.edu

Web: <https://dannycab.github.io>

Education

- **Georgia Institute of Technology** (Atlanta, GA)
Doctor of Philosophy in Physics, 2011
Thesis: **Evaluating and Extending a Novel Course Reform of Introductory Mechanics**
Advisor: Prof. Michael F. Schatz
[\[Online\]](#)
- **Georgia Institute of Technology** (Atlanta, GA)
Master of Science in Physics, 2007
- **University of Texas at Austin** (Austin, TX)
Bachelor of Science in Physics, 2004

Academic Experience

Positions

- **2025 – Present**, Lappan-Phillips Professor Math and Science Education, Department of Physics and Astronomy & Department of Computational Mathematics, Science, and Engineering, Michigan State University
- **2023 – 2025**, Lappan-Phillips Professor of Physics Education, Department of Physics and Astronomy & Department of Computational Mathematics, Science, and Engineering, Michigan State University
- **2020 – 2023**, Associate Professor, Department of Computational Mathematics, Science, and Engineering
- **2018 – 2023**, Lappan-Phillips Associate Professor of Physics Education, Department of Physics and Astronomy, Michigan State University
- **2017 – Present**, Adjunct Associate Professor, Department of Physics, University of Oslo
- **2013 – Present**, Affiliated Faculty, CREATE for STEM Institute, Michigan State University

- **2013 – 2018**, Assistant Professor, Department of Physics and Astronomy, Michigan State University
- **2011 – 2013**, Postdoctoral Researcher, Department of Physics, University of Colorado Boulder
- **2011 – 2013**, Research Affiliate, School of Physics, Georgia Institute of Technology
- **2005 – 2011**, Graduate Teaching and Research Assistant, School of Physics, Georgia Institute of Technology

Awards and Honors

- **2023** – Fellow of the American Physical Society, Group on Physics Education Research [\[Press\]](#)
“For foundational research and development on the roles of computation in physics education and contributions to research on undergraduate and graduate education in physics.”
- **2023** – APS Education Prize, Partnership for the Integration of Computing in Undergraduate Physics, Team Member [\[Press\]](#)
“For developing an active, inclusive, and supportive community of physics educators dedicated to integrating computation into their instruction; creating, reviewing, and disseminating instructional materials; and generating knowledge of computation in physics curricula and of effective practices.”
- **2022** – Physical Review PER Editor’s Suggestion (Topic: Computational Physics Education): *Student’ perspectives on computational challenges in physics class* [\[Paper\]](#)
- **2022** – Department Award for Improving Undergraduate Physics Education, Team Member, American Physical Society [\[Press\]](#)
- **2021** – Physical Review PER Editor’s Suggestion (Topic: Quantitative Methods for Education Research): *Framework for evaluating statistical models in physics education research* [\[Paper\]](#)
- **2021** – Physical Review PER Editor’s Suggestion (Topic: Graduate Physics Education): *Physics Graduate Record Exam does not help applicants “stand out”* [\[Paper\]](#)
- **2019** – Physical Review PER Editor’s Suggestion (Topic: Computational Physics Education): *Physics computational literacy: An exploratory case study using computational essays* [\[Paper\]](#)
- **2019** – Physics Education Research Conference Notable Paper [\[Press\]](#)
- **2019** – Featured in MSU Today for Teaching Innovation [\[Press\]](#)
- **2018** – President’s Distinguished Teaching Award, MSU [\[Press\]](#)
- **2018** – Teacher-Scholar Award, MSU [\[Press\]](#)
- **2017** – Featured in MSU Today for NSF grant [\[Press\]](#)
- **2016** – Physics Education Research Conference Notable Paper [\[Press\]](#)
- **2016** – College of Natural Science Teaching Prize, MSU
- **2015** – Thomas H. Osgood Memorial Awards for Faculty Excellence in Teaching, MSU
- **2014** – STEM Gateway Fellow, College of Natural Science, MSU

Publications

Citation Metrics



Marcos D Caballero ✎

Other names ▾

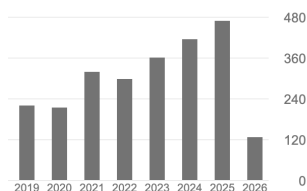
Professor, [Michigan State University](#)
Verified email at msu.edu - [Homepage](#)

[STEM Education](#) [Physics Education](#) [Computing Education](#)

FOLLOWING

Cited by [VIEW ALL](#)

	All	Since 2021
Citations	2985	1992
h-index	30	25
i10-index	62	51



☐ TITLE	CITED BY	YEAR
☐ Characterizing college science assessments: The three-dimensional learning assessment protocol JT Lavery, SM Underwood, RL Matz, LA Posey, JH Carmel, ... PloS one 11 (9), e0162333	290	2016
☐ Challenge faculty to transform STEM learning MM Cooper, MD Caballero, D Ebert-May, CL Fata-Hartley, SE Jardeleza, ... Science 350 (6258), 281-282	197	2015

Google Scholar citation count as of April 10, 2026

Key Metrics

- Citations (Lifetime): 2985
- Citations (2025): 468
- h-index (papers cited 30+ times): 30
- i10-index (papers cited 10+ times): 62

Writing in Review

1. Wang, Y., Renbarger, R., Boyd, T., Bolger, E., Finkelstein, N., Henderson, C., Caballero, M. D., Beach, A., & Simkins, S. (2026). *Facilitating systemic change in undergraduate engineering education: A literature meta-synthesis on change strategies*.
2. Bolger, E., Renbarger, R., Boyd, T., Wang, Y., Caballero, M. D., Finkelstein, N., Simkins, S., Henderson, C., & Beach, A. (2026). *Integrating machine learning into systematic literature reviews: Advancing a human-in-the-loop framework*.

Books, Edited Volumes, and Chapters

1. Caballero, M. D. (Ed.). (2025). *Integrating computing into the undergraduate physics curriculum*. IOP Publishing. <https://doi.org/10.1088/978-0-7503-5372-4>
 - Caballero, M. D. (2025). Integrating computing in physics courses [Book Chapter]. In *Integrating computing into the undergraduate physics curriculum* (pp. 1-1 to 1-14). IOP Publishing. <https://doi.org/10.1088/978-0-7503-5372-4ch1>
 - Caballero, M. D. (2025). Where can we go from here? [Book Chapter]. In *Integrating computing into the undergraduate physics curriculum* (pp. 14-1 to 14-17). IOP Publishing. <https://doi.org/10.1088/978-0-7503-5372-4ch14>

Invited Articles and Papers

1. Caballero, M. D., & Odden, T. O. B. (2024). Computing in physics education. *Nature Physics*, 20(3), 339–341. <https://doi.org/10.1038/s41567-023-02371-2>
2. Young, N. T., Tollefson, K., & Caballero, M. D. (2023). Making graduate admissions in physics more equitable. *Physics Today*, 76(7), 40–45. <https://doi.org/10.1063/PT.3.5271>
3. Odden, T. O. B., & Caballero, M. D. (2023). Physics Computational Literacy: What, Why, and How? In *The International Handbook of Physics Education Research: Learning Physics*. AIP Publishing LLC. https://doi.org/10.1063/9780735425477_019
4. Knaub, A. V., Aiken, J. M., & Caballero, M. D. (2019). Editorial: Focused Collection: Quantitative Methods in PER: A Critical Examination. *Physical Review Physics Education Research*, 15, 020001. <https://doi.org/10.1103/PhysRevPhysEducRes.15.020001>
5. Caballero, M. D., & Hjorth-Jensen, M. (2018). Integrating a Computational Perspective in Physics Courses. In S. Magazù (Ed.), *New Trends in Physics Education Research* (pp. 47–76). Nova Science Publishers.
6. Caballero, M. D. (2017). Taking A Scientific Approach To Physics Education. *Student Journal of Physics*, 6(1). https://www.iopb.res.in/~sjp/sjp_past_issues/V6N1/1.pdf
7. Caballero, M. D., Kohlmyer, M. A., & F., M. (2011). Fostering Computational Thinking. In N. S. Rebello, P. V. Engelhardt, & C. Singh (Eds.), *2011 Physics Education Research Conference Proceedings* (pp. 15–18). <https://doi.org/10.1063/1.3679982>
8. Bujak, K. R., Caballero, M. D., Schatz, M. F., Marr, M. J., & Catrambone, R. (2011). Comparing the Matter and Interactions Curriculum with a Traditional Physics Curriculum: A Think Aloud Study. *2011 AERA Conference Proceedings*. <https://arxiv.org/abs/1011.5449v1>

Conference Reports

1. Caballero, M. D., Engelhardt, L., Knaub, A. V., Kuchera, M., Puerto, M. L. del, Lunk, B., Roos, K., & Zimmerman, T. (2022). *2021 PICUP Virtual Capstone Conference Report* (R. C. Hilborn, Ed.). American Association of Physics Teachers. https://www.compadre.org/picup//events/pdfs/2021_PICUP_Capstone_Report_Final_Final_220502.pdf
2. Caballero, M. D., Engelhardt, L., Hilborn, R., Puerto, M. L. del, & Roos, K. (2019). PICUP: The Partnership for the Integration of Computation into Undergraduate Physics. *APS News*, 28(3). <https://www.aps.org/publications/apsnews/201903/backpage.cfm>
3. Caballero, M. D., Dounas-Frazer, D. R., Lewandowski, H. J., & Stetzer, M. R. (2018). Labs are Necessary, and We Need to Invest in Them. *APS News*, 27(5). <https://www.aps.org/publications/apsnews/201805/backpage.cfm>

Articles Appearing in Peer-Reviewed Journals

1. Stroupe, D., Byun, S., Christensen, J., Ortiz, N., Willison, J., Reinholz, D. L., & Caballero, M. D. (2026). Supporting teachers to integrate computational practices and design opportunities for equitable participation in science classrooms. *Cognition and Instruction*, 44(2), 110–152. <https://doi.org/10.1080/07370008.2026.2630727>
2. Byun, S., Brennan, L., Christensen, J., Ortiz, N., Shah, N., Reinholz, D., Stroupe, D., & Caballero, M. D. (2026). Equity-focused coaching: Negotiating teachers' interpretations of racialized and gendered participation patterns. *Journal of Teacher Education*, 77(3), 212–229. <https://doi.org/10.1080/00220272.2026.2120000>

[org/10.1177/00224871261417322](https://doi.org/10.1177/00224871261417322)

3. Sabo, H. C., Odden, T. O. B., & Caballero, M. D. (2026). How do we assess computation in physics? *The Physics Teacher*, 64(3), 200–203. <https://doi.org/10.1119/5.0174333>
4. Bolger, E., Nwobi, M., & Caballero, M. D. (2025). Characterizing faculty online learning community interactions using social network analysis. *Phys. Rev. Phys. Educ. Res.*, 21, 020133. <https://doi.org/10.1103/n3yf-5h29>
5. Geschwind, G., Vignal, M., Caballero, M. D., & Lewandowski, H. J. (2024). Evidence for validity and reliability of a research-based assessment instrument on measurement uncertainty. *Phys. Rev. Phys. Educ. Res.*, 20, 020125. <https://doi.org/10.1103/PhysRevPhysEducRes.20.020125>
6. Geschwind, G., Vignal, M., Caballero, M. D., & Lewandowski, H. J. (2024). Using a research-based assessment instrument to explore undergraduate students' proficiencies around measurement uncertainty in physics lab contexts. *Phys. Rev. Phys. Educ. Res.*, 20, 020105. <https://doi.org/10.1103/PhysRevPhysEducRes.20.020105>
7. Cooper, M. M., Caballero, M. D., Carmel, J. H., Duffy, E. M., Ebert-May, D., Fata-Hartley, C. L., Herington, D. G., Lavery, J. T., Nelson, P. C., Posey, L. A., Stoltzfus, J. R., Stowe, R. L., Sweeder, R. D., Tessmer, S., & Underwood, S. M. (2024). Beyond active learning: Using 3-dimensional learning to create scientifically authentic, student-centered classrooms. *Plos One*, 19(5), e0295887. <https://doi.org/10.1371/journal.pone.0295887>
8. Vignal, M., Geschwind, G., Pollard, B., Henderson, R., Caballero, M. D., & Lewandowski, H. J. (2023). Survey of physics reasoning on uncertainty concepts in experiments: An assessment of measurement uncertainty for introductory physics labs. *Phys. Rev. Phys. Educ. Res.*, 19, 020139. <https://doi.org/10.1103/PhysRevPhysEducRes.19.020139>
9. Young, N. T., Verboncoeur, N., Lam, D. C., & Caballero, M. D. (2023). Rubric-based holistic review represents a change from traditional graduate admissions approaches in physics. *Phys. Rev. Phys. Educ. Res.*, 19, 010134. <https://doi.org/10.1103/PhysRevPhysEducRes.19.010134>
10. Young, N. T., Tollefson, K., Zegers, R. G. T., & Caballero, M. D. (2022). Rubric-based holistic review: A promising route to equitable graduate admissions in physics. *Phys. Rev. Phys. Educ. Res.*, 18, 020140. <https://doi.org/10.1103/PhysRevPhysEducRes.18.020140>
11. Hamerski, P. C., McPadden, D., Caballero, M. D., & Irving, P. W. (2022). Students' perspectives on computational challenges in physics class. *Phys. Rev. Phys. Educ. Res.*, 18, 020109. <https://doi.org/10.1103/PhysRevPhysEducRes.18.020109>
12. Weller, D. P., Bott, T. E., Caballero, M. D., & Irving, P. W. (2022). Development and illustration of a framework for computational thinking practices in introductory physics. *Phys. Rev. Phys. Educ. Res.*, 18, 020106. <https://doi.org/10.1103/PhysRevPhysEducRes.18.020106>
13. Wilson, J., Pollard, B., Aiken, J. M., Caballero, M. D., & Lewandowski, H. J. (2022). Classification of open-ended responses to a research-based assessment using natural language processing. *Physical Review Physics Education Research*. <https://doi.org/10.1103/PhysRevPhysEducRes.18.010141>
14. Sand, O. P., Caballero, M. D., Mørken, K. M., & Lockwood, E. (2022). Three Cases That Demonstrate How Students Connect the Domains of Mathematics and Computing. *Journal of Mathematical Behavior*. <https://doi.org/10.1016/j.jmathb.2022.100955>
15. Sand, O. P., Lockwood, E., Caballero, M. D., & Mørken, K. M. (2022). Students' Development of a Logarithm Function in Python Using Taylor Expansions: A Teaching Design Case Study. *Digital Experiences in Mathematics Education*. <https://doi.org/10.1007/s40751-022-00104-3>

16. Pollard, B., Hobbs, R., Henderson, R., Caballero, M. D., & Lewandowski, H. J. (2021). Introductory physics lab instructors' perspectives on measurement uncertainty. *Physical Review Physics Education Research*, 17, 010133. <https://doi.org/10.1103/PhysRevPhysEducRes.17.010133>
17. Young, N. T., & Caballero, M. D. (2021). Predictive and Explanatory Models Might Miss Informative Features in Educational Data. *Journal of Educational Data Mining*, 13(4), 31–86. <https://doi.org/10.5281/zenodo.5806830>
18. Young, N. T., & Caballero, M. D. (2021). Physics Graduate Record Exam does not help applicants “stand out”. *Physical Review Physics Education Research*, 17, 010144. <https://doi.org/10.1103/PhysRevPhysEducRes.17.010144>
19. Aiken, J. M., De Bin, R., Lewandowski, H. J., & Caballero, M. D. (2021). Framework for evaluating statistical models in physics education research. *Physical Review Physics Education Research*, 17, 020104. <https://doi.org/10.1103/PhysRevPhysEducRes.17.020104>
20. Mikkelsen, N. J., Young, N. T., & Caballero, M. D. (2021). Investigating institutional influence on graduate program admissions by modeling physics graduate record examination cutoff scores. *Physical Review Physics Education Research*, 17, 010109. <https://doi.org/10.1103/PhysRevPhysEducRes.17.010109>
21. Aiken, J. M., De Bin, R., Hjorth-Jensen, M., & Caballero, M. D. (2020). Predicting time to graduation at a large enrollment american university. *PLOS ONE*, 15(11). <https://doi.org/10.1371/journal.pone.0242334>
22. Irving, P. W., McPadden, D., & Caballero, M. D. (2020). Communities of practice as a curriculum design theory in an introductory physics class for engineers. *Physical Review Physics Education Research*, 16, 020143. <https://doi.org/10.1103/PhysRevPhysEducRes.16.020143>
23. Odden, T. O. B., Marin, A., & Caballero, M. D. (2020). Thematic analysis of 18 years of physics education research conference proceedings using natural language processing. *Physical Review Physics Education Research*, 16, 010142. <https://doi.org/10.1103/PhysRevPhysEducRes.16.010142>
24. Pawlak, A., Irving, P. W., & Caballero, M. D. (2020). Learning assistant approaches to teaching computational physics problems in a problem-based learning course. *Physical Review Physics Education Research*, 16, 010139. <https://doi.org/10.1103/PhysRevPhysEducRes.16.010139>
25. Bain, K., Matz, R. L., Fata-Hartley, C. L., Caballero, M. D., Ebert-May, D., Underwood, S. M., Carmel, J. H., Herrington, D. G., Laverty, J. T., Duffy, E. M., Stoltzfus, J. R., Bender, L., Posey, L. A., Urban-Lurain, M., Stowe, R. L., Sweeder, R. D., Tessmer, S. H., & Cooper, M. M. (2020). Characterizing College Science Instruction: The Three-Dimensional Learning Observation Protocol. *PLOS ONE*, 15(6), e0234640. <https://doi.org/10.1371/journal.pone.0234640>
26. Odden, T. O. B., Lockwood, E., & Caballero, M. D. (2019). Physics computational literacy: An exploratory case study using computational essays. *Physical Review Physics Education Research*, 15, 020152. <https://doi.org/10.1103/PhysRevPhysEducRes.15.020152>
27. Funkhouser, K., Martinez, W., Henderson, R., & Caballero, M. D. (2019). Design, Analysis, Tools, and Apprenticeship (DATA) Lab. *European Journal of Physics*, 40(6), 065701. <https://doi.org/10.1088/1361-6404/ab2f0d>
28. Caballero, M. D., Chonacky, N., Engelhardt, L., Hilborn, R. C., Lopez del Puerto, M., & Roos, K. R. (2019). PICUP: A Community of Teachers Integrating Computation into Undergraduate Physics Courses. *The Physics Teacher*, 57(6), 397–399. <https://doi.org/10.1119/1.5124281>
29. Aiken, J. M., Henderson, R., & Caballero, M. D. (2019). Modeling student pathways in a physics bachelor's degree program. *Physical Review Physics Education Research*, 15, 010128. <https://doi.org/10.1103/PhysRevPhysEducRes.15.010128>

[//doi.org/10.1103/PhysRevPhysEducRes.15.010128](https://doi.org/10.1103/PhysRevPhysEducRes.15.010128)

30. Young, N. T., Allen, G., Aiken, J. M., Henderson, R., & Caballero, M. D. (2019). Identifying features predictive of faculty integrating computation into physics courses. *Physical Review Physics Education Research*, 15, 010114. <https://doi.org/10.1103/PhysRevPhysEducRes.15.010114>
31. Caballero, M. D., & Merner, L. (2018). Prevalence and nature of computational instruction in undergraduate physics programs across the united states. *Physical Review Physics Education Research*, 14, 020129. <https://doi.org/10.1103/PhysRevPhysEducRes.14.020129>
32. Matz, R. L., Fata-Hartley, C. L., Posey, L. A., Laverty, J. T., Underwood, S. M., Carmel, J. H., Herrington, D. G., Stowe, R. L., Caballero, M. D., Ebert-May, D., & Cooper, M. M. (2018). Evaluating the extent of a large-scale transformation in gateway science courses. *Science Advances*, 4(10). <https://doi.org/10.1126/sciadv.aau0554>
33. Stroupe, D., Caballero, M. D., & White, P. (2018). Fostering students' epistemic agency through the co-configuration of moth research. *Science Education*, 1–25. <https://doi.org/10.1002/sce.21469>
34. Laverty, J. T., & Caballero, M. D. (2018). Analysis of the most common concept inventories in physics: What are we assessing? *Physical Review Physics Education Research*, 14, 010123. <https://doi.org/10.1103/PhysRevPhysEducRes.14.010123>
35. Pawlak, A., Irving, P. W., & Caballero, M. D. (2018). Development of the Modes of Collaboration framework. *Physical Review Physics Education Research*, 14, 010101. <https://doi.org/10.1103/PhysRevPhysEducRes.14.010101>
36. Irving, P. W., Obsniuk, M. J., & Caballero, M. D. (2017). P³: a practice focused learning environment. *European Journal of Physics*, 38(5), 055701. <https://doi.org/10.1088/1361-6404/aa7529>
37. Caballero, M. D., Doughty, L., Turnbull, A. M., Pepper, R. E., & Pollock, S. J. (2017). Assessing learning outcomes in middle-division classical mechanics: The colorado classical mechanics and math methods instrument. *Physical Review Physics Education Research*, 13, 010118. <https://doi.org/10.1103/PhysRevPhysEducRes.13.010118>
38. Laverty, J. T., Underwood, S. M., Matz, R. L., Posey, L. A., Carmel, J. H., Caballero, M. D., Fata-Hartley, C. L., Ebert-May, D., Jardeleza, S. E., & Cooper, M. M. (2016). Characterizing College Science Assessments: The Three-Dimensional Learning Assessment Protocol. *PLOS ONE*, 11(9), e0162333. <https://doi.org/10.1371/journal.pone.0162333>
39. Cooper, M. M., Caballero, M. D., Ebert-May, D., Fata-Hartley, C. L., Jardeleza, S. E., Krajcik, J. S., Laverty, J. T., Matz, R. L., Posey, L. A., & Underwood, S. M. (2015). Challenge faculty to transform STEM learning. *Science*, 350(6258), 281–282. <https://doi.org/10.1126/science.aab0933>
40. Chasteen, S. V., Wilcox, B., Caballero, M. D., Perkins, K. K., Pollock, S. J., & Wieman, C. E. (2015). Educational transformation in upper-division physics: The Science Education Initiative model, outcomes, and lessons learned. *Physical Review Special Topics – Physics Education Research*, 11, 020110. <https://doi.org/10.1103/PhysRevSTPER.11.020110>
41. Wilcox, B. R., Caballero, M. D., Baily, C., Sadaghiani, H., Chasteen, S. V., Ryan, Q. X., & Pollock, S. J. (2015). Development and uses of upper-division conceptual assessments. *Physical Review Special Topics – Physics Education Research*, 11, 020115. <https://doi.org/10.1103/PhysRevSTPER.11.020115>
42. Caballero, M. D., Wilcox, B. R., Doughty, L., & Pollock, S. J. (2015). Unpacking students' use of mathematics in upper-division physics: Where do we go from here? *European Journal of Physics*, 36(6), 065004. <https://doi.org/10.1088/0143-0807/36/6/065004>

43. Ding, L., & Caballero, M. D. (2014). Uncovering the hidden meaning of cross-curriculum comparison results on the force concept inventory. *Physical Review Special Topics – Physics Education Research*, 10, 020125. <https://doi.org/10.1103/PhysRevSTPER.10.020125>
44. Hoskinson, A.-M., Couch, B. A., Zwickl, B. M., Hinko, K. A., & Caballero, M. D. (2014). Bridging physics and biology teaching through modeling. *American Journal of Physics*, 82(5), 434–441. <https://doi.org/10.1119/1.4870502>
45. Caballero, M. D., & Pollock, S. J. (2014). A model for incorporating computation without changing the course: An example from middle-division classical mechanics. *American Journal of Physics*, 82(3), 231–237. <https://doi.org/10.1119/1.4837437>
46. Caballero, M. D., Burk, J. B., Aiken, J. M., Thoms, B. D., Douglas, S. S., Scanlon, E. M., & Schatz, M. F. (2014). Integrating Numerical Computation into the Modeling Instruction Curriculum. *The Physics Teacher*, 52(1), 38–42. <https://doi.org/10.1119/1.4849153>
47. Wilcox, B. R., Caballero, M. D., Rehn, D. A., & Pollock, S. J. (2013). Analytic framework for students' use of mathematics in upper-division physics. *Physical Review Special Topics – Physics Education Research*, 9, 020119. <https://doi.org/10.1103/PhysRevSTPER.9.020119>
48. Hoskinson, A.-M., Caballero, M. D., & Knight, J. K. (2013). How can we improve problem solving in undergraduate biology?: Applying lessons from 30 years of physics education research. *Cell Biology Education - Life Science Education*, 12(2), 153–161. <https://doi.org/10.1187/cbe.12-09-0149>
49. Chasteen, S. V., Pepper, R. E., Caballero, M. D., Pollock, S. J., & Perkins, K. K. (2012). Colorado upper-division electrostatics diagnostic: A conceptual assessment for the junior level. *Physical Review Special Topics – Physics Education Research*, 8, 020108. <https://doi.org/10.1103/PhysRevSTPER.8.020108>
50. Caballero, M. D., Kohlmyer, M. A., & Schatz, M. F. (2012). Implementing and assessing computational modeling in introductory mechanics. *Physical Review Special Topics – Physics Education Research*, 8, 020106. <https://doi.org/10.1103/PhysRevSTPER.8.020106>
51. Caballero, M. D., Greco, E. F., Murray, E. R., Bujak, K. R., Marr, M. J., Catrambone, R., Kohlmyer, M. A., & Schatz, M. F. (2012). Comparing large lecture mechanics curricula using the Force Concept Inventory: A five thousand student study. *American Journal of Physics*, 80(7), 638–644. <https://doi.org/10.1119/1.3703517>
52. Kohlmyer, M. A., Caballero, M. D., Catrambone, R., Chabay, R. W., Ding, L., Haugan, M. P., Marr, M. J., Sherwood, B. A., & Schatz, M. F. (2009). Tale of two curricula: The performance of 2000 students in introductory electromagnetism. *Physical Review Special Topics – Physics Education Research*, 5, 020105. <https://doi.org/10.1103/PhysRevSTPER.5.020105>
53. Cochran, A. L., Barker, E. S., Caballero, M. D., & György-Ries, J. (2009). Placing the Deep Impact Mission into context: Two decades of observations of 9P/Tempel 1 from McDonald Observatory. *Icarus*, 199(1), 119–128. <https://doi.org/10.1016/j.icarus.2008.08.015>

Articles Appearing in Peer-Reviewed Conference Proceedings

1. Renbarger, R., Wang, Y., Boyd, T., Bolger, E., Ison, M., Finkelstein, N., Henderson, C., Beach, A., Simkins, S., & Caballero, M. D. (2026). ECR: Creating transformational STEM education: A meta-synthesis mapping a decade of systemic change research. *Proceedings of the American Society for Engineering Education Annual Conference*.
2. Wang, Y., Renbarger, R., Boyd, T., Bolger, E., Finkelstein, N., Henderson, C., Caballero, M. D.,

- Beach, A., & Simkins, S. (2026). Change strategies in undergraduate STEM education: A meta-synthesis and update of a decade of research. *Proceedings of the American Educational Research Association Annual Meeting*.
3. Renbarger, R., Wang, Y., Boyd, T., Bolger, E., Finkelstein, N., Henderson, C., Caballero, M. D., Beach, A., & Simkins, S. (2026). Making equitable change: A meta-synthesis of instructional change efforts for marginalized student groups. *Proceedings of the American Educational Research Association Annual Meeting*.
 4. Wang, Y., Bolger, E., Renbarger, R., Boyd, T., Finkelstein, N., Henderson, C., Beach, A., Simkins, S., & Caballero, M. D. (2026). ECR: Facilitating change in undergraduate STEM: A multidisciplinary, multimethod, meta-synthesis mapping a decade of growth. *Proceedings of the American Society for Engineering Education Annual Conference*.
 5. Chitchyan, S., Caballero, M. D., Nomer, M., & Karam, L. (2026). Feedback engagement and analysis in assessments (FEAST) in computing education. *Proceedings of the 57th ACM Technical Symposium on Computer Science Education v.2*, 1265–1266. <https://doi.org/10.1145/3770761.3777263>
 6. Bolger, E., Wang, Y., Renbarger, R., Caballero, M. D., Nwobi, M., Lem, C., Finkelstein, N., Simkins, S., Boyd, T., Henderson, C., & Beach, A. (2025). Integrating machine learning into systematic reviews: Strengthening evidence synthesis. *Proceedings of the 2025 Annual Meeting of the American Educational Research Association*.
 7. Bolger, E., & Caballero, M. (2025). Exploring the Impact of Unsupervised Clustering Methods in Systematic Literature Reviews. *Proceedings of the 56th ACM Technical Symposium on Computer Science Education v. 2*, 1391–1392. <https://doi.org/10.1145/3641555.3705184>
 8. Bolger, E., & Caballero, M. (2024). Using natural language processing to explore instructional change strategies in undergraduate science education literature. *Proceedings of the 55th ACM Technical Symposium on Computer Science Education v. 2*, 1930–1930. <https://doi.org/10.1145/3626253.3635341>
 9. Silvia, D. W., Caballero, M. D., Finzell, T., Frisbie, R., Hamerski, P., Bolger, E., Castle, S., Roca, R., & Tourangeau, P. (2023). Computing in support of disciplinary learning. *Proceedings of the 54th ACM Technical Symposium on Computer Science Education v. 2*, 1247–1247. <https://doi.org/10.1145/3545947.3573341>
 10. Hamerski, P. C., Silvia, D., & Caballero, M. D. (2022). Exploring self-efficacy in data science. *Proceedings of the 27th ACM Conference on Innovation and Technology in Computer Science Education Vol. 2*, 633–634. <https://doi.org/10.1145/3502717.3532131>
 11. Willison, J., Christensen, J., Byun, S., Stroupe, D., & Caballero, M. D. (2022). How do you eat an elephant? How problem solving informs computational instruction in high school physics. *Physics Education Research Conference 2022*, 494–499.
 12. Stroupe, D., Byun, S., Willson, J., Christensen, J., Caballero, M. D., & Reinholz, D. L. (2022). Teachers' use of resources for equitable integration of computation in science classrooms. In C. Chinn, E. Tan, C. Chan, & Y. Kali (Eds.), *Proceedings of the 16th international conference of the learning sciences-ICLS2022* (pp. 905–909). International Society of the Learning Sciences.
 13. Mackessy, G., Irving, P. W., Caballero, M. D., & Doughty, L. (2021). Comparing student conceptions and construction of while loops in modeling motion. In M. B. Bennett, B. W. Frank, & R. E. Vieyra (Eds.), *2021 Physics Education Research Conference Proceedings* (pp. 245–250). <https://doi.org/10.1119/perc.2021.pr.Mackessy>
 14. Waterson, A. C., Henderson, R., & Caballero, M. D. (2021). Analyzing time-to-degree for transfer

- students at a Large Midwestern University. In M. B. Bennett, B. W. Frank, & R. E. Vieyra (Eds.), *2021 Physics Education Research Conference Proceedings* (pp. 438–443). <https://doi.org/10.1119/perc.2021.pr.Watsonson>
15. Henderson, R., Funkhouser, K., & Caballero, M. D. (2019). A Longitudinal Exploration of Students' Beliefs about Experimental Physics. In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 214–219). <https://doi.org/10.1119/perc.2019.pr.Henderson>
 16. Young, N. T., & Caballero, M. D. (2019). Using machine learning to understand physics graduate school admissions. In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 669–674). <https://doi.org/10.1119/perc.2019.pr.Young>
 17. Odden, T. O. B., & Caballero, M. D. (2019). Computational Essays and Computational Literacy at the University of Oslo. In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 429–434). <https://doi.org/10.1119/perc.2019.pr.Odden>
 18. Weller, D., Caballero, M. D., & Irving, P. W. (2019). Investigating Teacher Learning Goals Involving Computation in High School Physics. In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 627–632). <https://doi.org/10.1119/perc.2019.pr.Weller>
 19. Bott, T., Weller, D., Caballero, M. D., & Irving, P. W. (2019). Preliminary Analysis of Student-Identified Themes around Computation in High School Physics. In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 57–62). <https://doi.org/10.1119/perc.2019.pr.Bott>
 20. Bumler, J., Hamerski, P. C., Caballero, M. D., & Irving, P. W. (2019). How do previous coding experiences influence undergraduate physics students? In Y. Cao, S. Wolf, & M. B. Bennett (Eds.), *2019 Physics Education Research Conference Proceedings* (pp. 69–74). <https://doi.org/10.1119/perc.2019.pr.Bumler>
 21. McPadden, D., Hamerski, P. C., Caballero, M. D., & Irving, P. W. (2018). Feedback as a mechanism for improving students scientific communication skills. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.McPadden>
 22. Sand, O. P., Odden, T. O. B., Lindstrøm, C., & Caballero, M. D. (2018). How computation can facilitate sensemaking about physics: A case study. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.Sand>
 23. Funkhouser, K., Caballero, M. D., Irving, P. W., & Sawtelle, V. (2018). What counts in laboratories: Toward a practice-based identity survey. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.Funkhouser>
 24. Leary, A., Irving, P. W., & Caballero, M. D. (2018). The difficulties associated with integrating computation into undergraduate physics. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.Leary>
 25. Griswold, K., McPadden, D., Caballero, M. D., & Irving, P. W. (2018). Denoting and Comparing Leadership Attributes and Behaviors in Group Work. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.Griswold>
 26. Solli, R., Aiken, J. M., Henderson, R., & Caballero, M. D. (2018). Examining the relationship be-

- tween student performance and video interactions. In A. Traxler, Y. Cao, & S. Wolf (Eds.), *2018 Physics Education Research Conference Proceedings*. <https://doi.org/10.1119/perc.2018.pr.Solli>
27. Irving, P. W., & Caballero, M. D. (2017). Expanding the PICUP community of practice. In L. Ding, A. Traxler, & Y. Cao (Eds.), *2017 Physics Education Research Conference Proceedings* (pp. 188–191). <https://doi.org/10.1119/perc.2017.pr.042>
 28. Hawkins, N., Obsniuk, M. J., Irving, P. W., & Caballero, M. D. (2017). Examining Thematic Variation in a Phenomenographical Study on Computational Physics. In L. Ding, A. Traxler, & Y. Cao (Eds.), *2017 Physics Education Research Conference Proceedings* (pp. 168–171). <https://doi.org/10.1119/perc.2017.pr.037>
 29. Aiken, J. M., & Caballero, M. D. (2016). Methods for analyzing pathways through a physics major. In D. L. Jones, L. Ding, & A. Traxler (Eds.), *2016 Physics Education Research Conference Proceedings* (pp. 28–31). <https://doi.org/10.1119/perc.2016.pr.002>
 30. Caballero, M. D. (2015). Computation across the curriculum: What skills are needed? In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 79–82). <https://doi.org/10.1119/perc.2015.pr.015>
 31. Irving, P. W., Sawtelle, V., & Caballero, M. D. (2015). Troubleshooting Formative Feedback in P³ (A group-based learning environment). In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 155–158). <https://doi.org/10.1119/perc.2015.pr.034>
 32. Laverty, J. T., Cooper, M. M., & Caballero, M. D. (2015). Developing the Next Generation of Physics Assessments. In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 187–190). <https://doi.org/10.1119/perc.2015.pr.042>
 33. Pawlak, A., Irving, P. W., & Caballero, M. D. (2015). Identification of a shared answer-making game in group context. In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 255–258). <https://doi.org/10.1119/perc.2015.pr.059>
 34. Obsniuk, M. J., Irving, P. W., & Caballero, M. D. (2015). A Case Study: Novel Group Interactions through Computational Physics. In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 239–242). <https://doi.org/10.1119/perc.2015.pr.055>
 35. Turnbull, A., Doughty, L., Sawtelle, V., & Caballero, M. D. (2015). Student Ideas around Vector Decomposition in the Upper-Division. In A. D. Churukian, D. L. Jones, & L. Ding (Eds.), *2015 Physics Education Research Conference Proceedings* (pp. 239–242). <https://doi.org/10.1119/perc.2015.pr.079>
 36. Doughty, L., & Caballero, M. D. (2014). Rubric Design for Separating the Roles of Open-Ended Assessments. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 71–74). <https://doi.org/10.1119/perc.2014.pr.014>
 37. Laverty, J. T., Tessmer, S. H., Cooper, M. M., & Caballero, M. D. (2014). Engaging Physics Faculty in Course Transformation. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 147–150). <https://doi.org/10.1119/perc.2014.pr.033>
 38. F., Wolf. S., Doughty, L., Irving, P. W., Sayre, E. C., & Caballero, M. D. (2014). Just Math: A new

- epistemic frame. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 275–278). <https://doi.org/10.1119/perc.2014.pr.065>
39. Aiken, J. M., Lin, S.-Y., Douglas, S. S., Greco, E. F., Thoms, B. D., Caballero, M. D., & Schatz, M. F. (2014). Student Use of a Single Lecture Video in a Flipped Introductory Mechanics Course. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 19–22). <https://doi.org/10.1119/perc.2014.pr.001>
 40. Douglas, S. S., Lin, S.-Y., Aiken, J. M., Greco, E. F., Thoms, B. D., Caballero, M. D., & Schatz, M. F. (2014). Peer Evaluation of Video Lab Reports in a Blended Introductory Physics Course. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 75–78). <https://doi.org/10.1119/perc.2014.pr.015>
 41. Lin, S.-Y., Douglas, S. S., Aiken, J. M., Greco, E. F., Thoms, B. D., Caballero, M. D., & Schatz, M. F. (2014). Peer Evaluation of Video Lab Reports in an Introductory Physics MOOC. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2014 Physics Education Research Conference Proceedings* (pp. 163–166). <https://doi.org/10.1119/perc.2014.pr.037>
 42. Caballero, M. D., & Pollock, S. J. (2013). Assessing Student Learning in Middle-Division Classical Mechanics/Math Methods. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2013 Physics Education Research Conference Proceedings* (pp. 81–84). <https://doi.org/10.1119/perc.2013.pr.008>
 43. Aiken, J. M., Lin, S.-Y., Douglas, S. S., Greco, E. F., Thoms, B. D., Caballero, M. D., & Schatz, M. F. (2013). The Initial State of Students Taking an Introductory Physics MOOC. In P. V. Engelhardt, A. D. Churukian, & D. L. Jones (Eds.), *2013 Physics Education Research Conference Proceedings* (pp. 53–56). <https://doi.org/10.1119/perc.2013.pr.001>
 44. Caballero, M. D., Wilcox, B. R., Pepper, R. E., & Pollock, S. J. (2012). ACER: A Framework on the Use of Mathematics in Upper-division Physics. In P. V. Engelhardt, A. D. Churukian, & N. S. Rebello (Eds.), *2012 Physics Education Research Conference Proceedings* (pp. 90–93). <https://doi.org/10.1063/1.4789659>
 45. Wilcox, B. R., Caballero, M. D., Pepper, R. E., & Pollock, S. J. (2012). Upper-division Student Understanding of Coulomb's Law: Difficulties with Continuous Charge Distributions. In P. V. Engelhardt, A. D. Churukian, & N. S. Rebello (Eds.), *2012 Physics Education Research Conference Proceedings* (pp. 418–421). <https://doi.org/10.1063/1.4789741>
 46. Aiken, J. M., Caballero, M. D., Douglas, S. S., Burk, J. B., Scanlon, E. M., Thoms, B. D., & Schatz, M. F. (2012). Understanding Student Computational Thinking with Computational Modeling. In P. V. Engelhardt, A. D. Churukian, & N. S. Rebello (Eds.), *2012 Physics Education Research Conference Proceedings* (pp. 46–49). <https://doi.org/10.1063/1.4789648>